Time Series

Spring Semester 2026, EPFL
École Polytechnique Fédérale de Lausanne
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Practice Exam — Week 12: Non-Linear Time Series — ARCH & GARCH

Duration: 60 minutes · No calculator · Answer on paper

Name: ____________________ Surname: ____________________

SCIPER (Student Card Number): ____________________

Exercise	Points
1	
2	
3	
4	
Total	

Justify every answer. Throughout, ε_t is a mean-zero white noise of variance σ^2.

Problem 1 (5 points)

(ARCH(2): conditional vs. unconditional moments and stationarity.) Let $\{r_t\}$ follow an ARCH(2) model

$$r_t = \sigma_t \eta_t, \qquad \sigma_t^2 = \beta_0 + \beta_1 r_{t-1}^2 + \beta_2 r_{t-2}^2,$$

with $\beta_0 > 0,\ \beta_1, \beta_2 \geq 0$, where $\{\eta_t\}$ is a white noise of mean 0 and *unit* variance, independent of $\mathcal{F}_{t-1}$ (the information up to time $t-1$). Here σ_t is $\mathcal{F}_{t-1}$-measurable. Take the numerical values $\beta_0 = 0.2$, $\beta_1 = 0.3$, $\beta_2 = 0.5$.

(a) Using the law of iterated expectation (condition on $\mathcal{F}_{t-1}$), show that $\mathbb{E}[r_t \mid \mathcal{F}_{t-1}] = 0$ and deduce $\mathbb{E}[r_t] = 0$. State also the conditional variance $\mathrm{Var}(r_t \mid \mathcal{F}_{t-1})$. (1.5 pts)

(b) Assuming $\{r_t\}$ is (weakly) stationary, use the law of total variance to derive the marginal variance $\mathrm{Var}(r_t)$ as a function of $\beta_0, \beta_1, \beta_2$, naming every probability law you use. Then give its numerical value for the parameters above. (2 pts)

(c) State the necessary condition on β_1, β_2 for a (positive, finite) stationary variance to exist, check it for the given numbers, and explain in one sentence why this is the $p = 2$, $q = 0$ instance of the general $\sum_j \alpha_j + \sum_j \beta_j < 1$ rule. (1.5 pts)

Problem 2 (5 points)

(GARCH is mean-zero white noise; stationarity & long-run variance.) A GARCH(m, r) process is

$$r_t = \sigma_t \eta_t, \qquad \sigma_t^2 = \alpha_0 + \sum_{j=1}^{m} \alpha_j r_{t-j}^2 + \sum_{j=1}^{r} \beta_j \sigma_{t-j}^2,$$

with $\alpha_0 > 0$, $\alpha_j, \beta_j \geq 0$, and $\{\eta_t\}$ white noise of mean 0 and unit variance, independent of $\mathcal{F}_{t-1}$; σ_t is $\mathcal{F}_{t-1}$-measurable.

(a) Prove that $\mathbb{E}[r_t] = 0$ and that, for every lag $\tau \neq 0$, $\mathrm{Cov}(r_{t+\tau}, r_t) = 0$, hence $\mathrm{Corr}(r_{t+\tau}, r_t) = 0$. (*Hint:* for $\tau > 0$ condition on $\mathcal{F}_{t+\tau-1}$ and pull out the measurable factors $r_t, \sigma_{t+\tau}$.) Conclude that $\{r_t\}$ is white noise. (2.5 pts)

(b) For a stationary GARCH, derive the unconditional variance $\mathrm{Var}(r_t) = \alpha_0 / \big(1 - \sum_j \alpha_j - \sum_j \beta_j\big)$ by taking unconditional expectations of the σ_t^2 recursion. State the resulting weak-stationarity condition. (*You may use $\mathbb{E}[r_{t-j}^2] = \mathbb{E}[\sigma_{t-j}^2] = \mathrm{Var}(r_t)$ at stationarity; justify this identity briefly.*) (1.5 pts)

(c) A GARCH$(1, 1)$ is fitted to daily log-returns with $\alpha_0 = 2 \times 10^{-5}$, $\alpha_1 = 0.10$, $\beta_1 = 0.85$. Verify weak stationarity, compute the unconditional variance and the implied daily volatility $\sqrt{\mathrm{Var}(r_t)}$, and comment in one sentence on the persistence $\alpha_1 + \beta_1$. (1 pt)

Problem 3 (5 points)

(ARCH(1): kurtosis, the ACF of the squares, and a volatility forecast.) Let $\{r_t\}$ be a stationary ARCH(1),

$$r_t = \sigma_t \eta_t, \qquad \sigma_t^2 = \alpha_0 + \alpha_1 r_{t-1}^2,$$

with $\alpha_0 > 0$, $0 < \alpha_1 < 1$, and *Gaussian* innovations $\eta_t \sim \mathcal{N}(0,1)$ independent of $\mathcal{F}_{t-1}$. Recall $\mathbb{E}[r_t] = 0$ and $\mathrm{Var}(r_t) = \alpha_0/(1-\alpha_1)$. You may use $\mathbb{E}[\eta_t^4] = 3$.

(a) Prove that for $\tau > 0$,

$$\mathrm{Cov}\big(r_{t+\tau}^2, r_t^2\big) = \alpha_1 \,\mathrm{Cov}\big(r_{t+\tau-1}^2, r_t^2\big),$$

by conditioning on $\mathcal{F}_{t+\tau-1}$ and using $\mathbb{E}[\eta^2 \mid \mathcal{F}_{t+\tau-1}] = 1$; be sure to show the constant terms cancel using $\mathbb{E}[r_t^2] = \alpha_0/(1-\alpha_1)$. Deduce that, provided $0 < \alpha_1 < 1/\sqrt{3}$, $\mathrm{Corr}(r_{t+\tau}^2, r_t^2) = \alpha_1^{|\tau|}$. (2.5 pts)

(b) The kurtosis of an ARCH(1) is $\kappa = \dfrac{3(1-\alpha_1^2)}{1-3\alpha_1^2}$ (valid when $\alpha_1^2 < 1/3$). An estimated model has $\alpha_1 = 0.5$. Compute κ, comment on the tails versus the normal, and report $\mathrm{Corr}(r_{t+\tau}^2, r_t^2)$ at lags $\tau = 1, 2, 3$. Why does the constraint for a finite fourth moment ($\alpha_1^2 < 1/3$) differ from the one for a finite variance ($\alpha_1 < 1$)? (1.5 pts)

(c) For the same model take $\alpha_0 = 4 \times 10^{-4}$ and $\alpha_1 = 0.5$, and suppose the latest observed return is $r_t = 0.04$. Give the one-step-ahead conditional variance σ_{t+1}^2 and a 95% predictive interval for r_{t+1} under Gaussian innovations; compare its width with the unconditional $\pm 1.96\sqrt{\mathrm{Var}(r_t)}$ band and interpret. (1 pt)

Problem 4 (5 points)

(Revision of Week 11: diagnostics — Ljung–Box test and AIC/BIC selection.) A mean-zero ARMA$(1,1)$ model is fitted to $n = 100$ observations. The residual sample autocorrelations at lags 1–5 are

$$\hat{r}_1 = 0.05, \quad \hat{r}_2 = -0.12, \quad \hat{r}_3 = 0.08, \quad \hat{r}_4 = -0.10, \quad \hat{r}_5 = 0.06.$$

Recall the Ljung–Box statistic $Q_m = n(n+2)\sum_{\tau=1}^{m} \hat{r}_\tau^2/(n-\tau)$, which under $H_0 : \rho_1 = \cdots = \rho_m = 0$ is approximately χ^2_{m-p-q} for an ARMA(p,q) fit. Useful quantiles: $\chi^2_{2,0.95} = 5.991$, $\chi^2_{3,0.95} = 7.815$, $\chi^2_{4,0.95} = 9.488$.

(a) Compute the Ljung–Box statistic Q_5 (show the per-lag terms). (1.5 pts)

(b) Test $H_0 : \rho_1 = \cdots = \rho_5 = 0$ at the 5% level: state the degrees of freedom, the critical value, the decision, and the practical conclusion about model adequacy. Would the decision change had an AR(1) been fitted instead? (1.5 pts)

(c) Recall $\text{AIC} = -2\log L + 2k$ and $\text{BIC} = -2\log L + k\log n$ with $k = p+q+1$ for an ARMA(p,q). A competing AR(3) (i.e. $k = 4$) is compared with the ARMA$(1,1)$ (i.e. $k = 3$) at $n = 100$. By how much must $-2\log L$ drop for AIC, respectively BIC, to prefer the larger model, and which criterion is more willing to select it? Why is in-sample R^2 a poor tool for this comparison? (2 pts)